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INTERNATIONAL GCSE

MARK SCHEME

MAXIMUM MARK: 80

SYLLABUS/COMPONENT: 0625/03

PHYSICS
Paper 3 (Extended)

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1	**(a)**	**(i)**	Acceleration / increase in speed	**M1**	
			Uniform / constant or in a straight line	**A1**	
		(ii)	Uniform speed	**B1**	
			Velocity changes / motion in a circle / accelerates	**B1**	**4**
	(b)		Similarity: same value / 6m/s or velocity changing	**B1**	
			Difference: opposite directions / up at E, down at C	**B1**	**2**
	(c)	**(i)**	Average speed x time / area under graph / 3 x 20	**C1**	
			60 m	**A1**	
		(ii)	6 x 52	**C1**	
			312m	**A1**	**4**
					[10]
2	**(a)**		750 N	**A1**	**1**
	(b)		p.e. lost / converted = mgh or weight x height	**C1**	
			750 x 15 or 75 x10 x15 = 11250 (J)	**C1**	
			p.e. lost = k.e. gained = 11250 (J)	**A1**	**3**
	(c)		Any 3 of: heat in water / rock (kinetic) energy of (moved) water / to make water move/ make waves some k.e. still in (sinking) rock sound energy on impact / of splash	**B3**	**3**
			(just heat and sound **C1**)		**[7]**
3	**(a)**	**(i)**	Extension proportional to load however expressed	**B1**	
		(ii)	Any relevant arithmetic to show direct proportion (or straight line graph <u>with values</u>)	**B1**	**2**
	(b)	**(i)**	Work done = force x distance / 400 x 0.210	**C1**	
			84.0 J	**A1**	
		(ii)	(total) work/time or (24 x) 84/60 (apply e.c.f from **(i)**)	**C1**	
			33.6 W	**A1**	**4**
					[6]
4	**(a)**		Water molecules at higher temps. have higher (av) k.e. / energy	**B1**	
			Higher energy molecules (have greater chance to) escape the surface	**B1**	
			Higher energy molecules have energy to break liquid “bonds” or separate liquid molecules or more evaporation at 85°C (lowers level)	**B1**	**3**
	(b)		Heat for evaporation = 34 500 – 600 = (33 900)	**C1**	
			Sp. latent heat of evaporation = heat/mass evap. or 33 900 / 15	**C1**	
			2260 J/g (method and working correct, but no heat loss used, 2/3)	**A1**	
			(600 added or 34 500 used can score **2 max**)		**3**

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5	**(a)**	**(i)**	Thermopile / thermocouple / (blackened) thermometer / infra red detector or use ammeter / voltmeter in supply circuit	**B1**	
		(ii)	One of: same distance of plate to detector or use two identical detectors or same time (after switching on)	**B1**	
		(iii)	Dull black better radiator / radiates more than silver / or emits more heat / radiation	**B1**	
		(iv)	Infra red (i.r.)	**A1**	**4**
	(b)		any correct example e.g. heating water or chimney	**M1**	
			current clear and complete	**A1**	
			direction shown correctly by arrows	**A1**	**3**
					[7]
6	**(a)**	**(i)**	Refraction at Q approx. correct, ray emerge from AB parallel PQ	**B1**	
		(ii)	Angle of incidence correctly marked	**B1**	
			Angle of refraction correctly marked	**B1**	
			(can score even if incorrect / no refraction shown)		**3**
	(b)	**(i)**	Refractive index = speed in air / speed in glass	**B1**	
		(ii)	Refractive index = $(3 \times 10^8 / 2 \times 10^8) = 1.5$	**B1**	**2**
	(c)	**(i)**	Wavelength = v/f or $3 \times 10^8 / 6 \times 10^{14}$	**C1**	
			Wavelength = 5×10^{-7} m	**A1**	**2**
					[7]
7	**(a)**		C,R,C,R,C,R marked (or v.v.) along XY	**B1**	**1**
	(b)	**(i)**	Above normal / high air pressure or particles close together	**B1**	
		(ii)	Below normal / low pressure or particles further apart	**B1**	**2**
	(c)		Oscillation / vibration of particles / molecules (or particles / molecules move to and fro)	**B1**	
			Oscillation is along XY	**B1**	**2**
	(d)		Time = distance / speed or (2x) 50/340	**C1**	
			Time = 0.29 s	**A1**	**2**

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8	**(a)**		1.52 kW	**A1**	**1**
	(b)	**(i)**	Each appliance is connected across 240 V supply or equivalent	**B1**	
		(ii)	Any 2: all work on same voltage or on 240 V or mains OR all have full/stated power OR each can be on or off OR one goes off/breaks others stay on	**B2**	**3**
	(c)	**(i)**	Current = power/voltage or 200/240	**C1**	
			Current = 0.83 A	**A1**	
		(ii)	Energy = power x time or 1.2 x 3	**C1**	
			Energy = 3.6 kWh or 1.3 x 10^7 J	**A1**	
		(iii)	Current = 60/240	**C1**	
			R= V/I or 240/0.25	**C1**	
			R =960Ω	**A1**	**7**
					[11]
9	**(a)**		Solenoid ends connected to meter, both labelled	**B1**	
			One magnet in correct position to enter / leave solenoid, labelled	**B1**	**2**
	(b)		Push magnet into coil / pull out / move near end of coil	**B1**	**1**
	(c)		(magnet has / produces) magnetic lines of force / magnetic field	**B1**	
			lines cut (coils of) solenoid / coils / wires	**B1**	**2**
	(d)	**(i)**	Pull magnet out of coil / reverse effect to answer **(b)**	**B1**	
		(ii)	Move magnet faster or effect in **(a)** faster	**B1**	**2**
					[7]
10	**(a)**		Analogue, continuously increasing / decreasing readings	**B1**	
			Digital, readings increase / decrease by one unit	**B1**	**2**
	(b)	**(i)**	Transistors + other components such as resistors	**B1**	
		(ii)	Standard symbol, must have labeled inputs and output	**B1**	
		(iii)	Both inputs 0 (off), or either one input 0 (off), output 0 (off)	**B1**	
			Both inputs 1 (on), output 1 (on) OR correct truth table drawn (**C1**) Some explanation of what truth table shows (**A1**)	**B1**	**4**
					[6]
11	**(a)**		Particle 1 carries straight on	**B1**	
			Particle 2 (slightly) deflected (less than 90°)	**B1**	
			Particle 3 “turns back” / (deflected more than 90°)	**B1**	**3**
	(b)		Nucleus is heavy /dense / all or most of mass in atom in nucleus	**B1**	
			Most of atom is space or nucleus is (very) small cf. atom	**B1**	**2**
	(c)		(mass) 4	**B1**	**1**
					[6]

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PAPER TOTAL = [80]